GM E41 (L5P Duramax) — Flash procedure in Good-Gravy-2 / V-OP

Document purpose: Summarize how the **E41** flash path is implemented in this repo (orchestrator + engine + ECU DB), where evidence came from, and **where to improve**.
Audience: Engineers extending flash to more ECUs.
Sources: `shared/ecuDatabase.ts`, `shared/pcanFlashOrchestrator.ts`, `client/src/lib/pcanFlashEngine.ts`, project notes (`todo.md`, BUSMASTER traces).

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1. Role of the E41 in the vehicle

- **E41** is the primary **engine control module** for **GM L5P Duramax** (6.6L diesel).
- **UDS on CAN** for flash uses **physical addressing** typically **TX `0x7E0` / RX `0x7E8`** (ISO 15765).
- **Calibration / DAQ** on some platforms may use **XCP** on other IDs (e.g. 1 Mbps) — flash in this app is modeled on **UDS/GMLAN** traces, not XCP.

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2. Where the behavior lives (code map)

Layer	File	Responsibility
ECU profile	`shared/ecuDatabase.ts`	`E41` entry: protocol `GMLAN`, `xferSize`, `seedLevel`, CAN IDs, `usesTransferExit`, patch/flash step lists
Flash plan (ordered steps, labels, timeouts)	`shared/pcanFlashOrchestrator.ts`	`generateFlashPlan()` — phases, GMLAN broadcast session, security, verification, key cycle, cleanup
Execution	`client/src/lib/pcanFlashEngine.ts`	Sends frames, handles **NRC 0x78**, **PriRC**, **TransferData**, **UUDT TesterPresent** keepalive
Security key	`client/src/lib/computeSecurityKeyClient.ts` / seed-key pipeline	GM AES-style keying for seed/key (profile-dependent)
Container binary	`shared/flashFileValidator.ts`, parsers	Block layout, `rc34`, lengths — must match DevProg / OEM tooling

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3. E41-specific configuration (`ECU_DATABASE.E41`)

From `shared/ecuDatabase.ts` (conceptual; verify in file for exact values):

- **`protocol: 'GMLAN'`** — Uses **functional broadcast** on **`0x101`** (UUDT `FE …`) for session entry, **not** a naive copy of "standard UDS only on `0x7E0`".
- **`txAddr: 0x7E0`, `rxAddr: 0x7E8`** — Physical ECM request/response.
- **`xferSize: 0xFFE`** — Matches BUSMASTER **RequestDownload** form **`34 00 00 0F FE`** (per-block format differs from some other GM ECUs using `0xFF8`).
- **`seedLevel: 0x01`** — SecurityAccess level for seed/key (paired with key level `+1` in orchestrator).

- `usesTransferExit: false` — Successful **E41** stock flash trace showed **no service** `0x37` across the log; next block's `0x34` implicitly continues. **Do not** force UDS-style `0x37` between blocks for GMLAN E41.
- `patchNecessary: true` — Patch pipeline may run before/around main flash per container strategy.

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4. High-level flash phases (orchestrator)

Phases are expressed as `FlashPhase` in `pcanFlashOrchestrator.ts`:
`PRE_CHECK` → `VOLTAGE_INIT` → `SESSION_OPEN` → `SECURITY_ACCESS` → `PRE_FLASH` / `BLOCK_TRANSFER` → `POST_FLASH` (often skipped GMLAN) → `VERIFICATION` → `KEY_CYCLE` → `CLEANUP`.

4.1 GMLAN vs UDS (tester present)

- **Standard UDS** ECUs: `0x3E 0x00` on physical address → expect `0x7E`.
- **GMLAN E41**: **USDT** `0x3E 0x00` **on ECM often yields NRC** `0x12` (sub-function not supported). **Keepalive** is **UUDT** on functional address `0x101`: `FE 01 3E` (cyclic, e.g. 500 ms), **after** the programming-mode steps that bring the ECU into bootloader — **not** before `0x28` **/** `A5` in the reference trace.

4.2 SESSION_OPEN — BUSMASTER-derived broadcast sequence (E41 / L5P)

Order and delays were tuned from a **successful stock flash** CAN log (BUSMASTER). Conceptual sequence on `0x101` (UUDT):

1. **ReturnToNormal** — `FE 01 20`
2. **ReadDID** `0xB0` (SW versions) — `FE 02 1A B0` (delay after RTN ~1000 ms in plan)
3. **DiagnosticSessionControl programming** — `FE 02 10 02`
4. **DisableNormalCommunication** — `FE 01 28`
5. **ReportProgrammedState** — `FE 01 A2`
6. **ProgrammingMode enable** — `FE 02 A5 01` (then ~1000 ms before next in trace)
7. **ProgrammingMode complete** — `FE 02 A5 03` → ECU transitions toward **bootloader**
8. **Start cyclic UUDT TesterPresent** — `FE 01 3E` (plan includes explicit first shot; engine runs interval)

Note: There is **no** separate **physical** `0x10 0x02` **on** `0x7E0` between broadcast and security in this reference — the functional programming session is the entry path.

4.3 SECURITY_ACCESS

- Physical: `0x27` seed at `seedLevel`, then key at `seedLevel+1`.
- **GMLAN:** Steps may be marked `nonFatal` if seed/key already succeeded in **PRE_CHECK** — post-bootloader seed can **time out** while flash still proceeds (engine safety net).

4.4 Erase and RequestDownload (blocks)

- **No standalone RoutineControl erase (** `0x31` **)** for GMLAN E41 — `0x31` has been observed as **NRC** `0x11` (service not supported). **Erase is implicit** in `0x34`: expect **NRC** `0x78` (response pending) while the ECU erases, then positive `0x74` when ready.
- **PriRC (bootloader entry):** Once per run, engine sends **SendCustomGMPriRC** — ISO-TP single frame `05 34 00 00 0F FE` after security, **before** the first per-block `0x34` from container `rc34` (see `sendGmBootloaderPriRc()` in `pcanFlashEngine.ts`).
- **Per block:** `0x34` → `0x36` TransferData chunks → **no** `0x37` for E41 (`usesTransferExit: false`).

4.5 POST_FLASH

- **RoutineControl** `0x31 0x01 0xFF 0x01` (check programming dependencies) is **omitted for GMLAN** in the plan (not supported or not used on typical GM ECMs in this flow).

4.6 VERIFICATION (after last transfer)

Reference order on physical + functional:

1. **ReturnToNormal** on `0x101` — `FE 01 20`
2. **Finalize** — `0xAE 0x28 0x80` on `0x7E0` (ECU may stay silent ~12 s — **reboot**)
3. **ReadDID** `0x90` (VIN), `0xC1` **…** `0xC6` (calibration IDs), `0xD0` (unlock / tuning status), `0xCC` (programming counter / CRC-style DID)
4. **ClearDTC** — `0x04` on `0x7DF` (GMLAN-style clear, not always same as UDS `0x14` framing)

4.7 KEY_CYCLE & CLEANUP

- **ECUReset** `0x11` on GMLAN E41 often **NRC** `0x11` — **ignition cycle** does the reset.
- User prompts: **KEY_OFF** → **KEY_ON** → **WAIT_BOOT**.
- **CLEANUP: ClearDTC**, **ReturnToNormal** on `0x101` — **no** reliance on `0x11 0x01` for E41.

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5. Evidence and parity

- **BUSMASTER** full-stock flash: timing, **absence of** `0x37`, **PriRC** form, **post-flash** `0xAE` sequence.
- **DevProg** (C#) is the **authoritative** container layout reference — TypeScript must stay aligned (`@devprog` rule).
- **Ford / other UDS** profiles use different `flashSequence` and UDS-only steps — compare `FORD_UDS_FLASH_SEQUENCE` vs `GM_FLASH_SEQUENCE` in `ecuDatabase.ts`.

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6. Improvement opportunities (E41-focused)

Area	Idea
Timing	Re-validate `delayBeforeMs` against newest GM logs (battery voltage, gateway load).
Security	Unlocked / HPTuners ECUs may accept **relaxed** keys — document test vs production ECU.
Verification	Harden DID reads if ECU is slow to boot after `0xAE` (retries, longer timeout).
Multi-block edge cases	Single-block "cal-only" containers that mark `OS=true` — orchestrator may warn and flash all blocks; confirm UX.
Telemetry	Structured flash logs for regression: compare frame counts per phase vs golden trace.

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7. Strategy: many ECUs, many procedures

Goal: Reusable **procedure** per ECU without copying unmaintainable one-off code.

1. **Data-driven ECU profile (`EcuConfig`)**
   - Protocol: `GMLAN` vs `UDS` (+ future: J1939, OEM-specific).
   - Addresses, `xferSize`, `usesTransferExit`, `seedLevel`, `patchNecessary`, `flashSequence` / `patchSequence`.
2. **Capture ground truth**
   - Successful flash **CAN log** (BUSMASTER, PCAN, etc.) + **DevProg** container for same strategy.

3. **Map deltas to branches**
   - If only **timing** differs: adjust `delayBeforeMs` / timeouts in **`generateFlashPlan`** or per-ECU overrides (new optional field on `EcuConfig` if needed).
   - If **services** differ: add `if (ecuType === 'X')` or prefer **`ecuConfig.flashHooks`** style flags (e.g. `skipTransferExit`, `postFlashVerificationDids`).
4. **Single execution path**
   - **`pcanFlashEngine`** should stay the **one** place for NRC `0x78`, PriRC, keepalive — avoid duplicating in bridge-only code.
5. **Regression tests**
   - Simulator / unit tests on **`generateFlashPlan`** for each ECU: expected command count, presence/absence of `0x37`, phase order.
6. **Documentation**
   - One **markdown + exported PDF** per "reference ECU" (like this document) when the procedure stabilizes.

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8. Revision history (manual)

Date	Note
2026-04-14	Generated from codebase review (orchestrator + ecuDatabase + flash engine).

End of document.